Data Structures (CS301)

Multiple Choice Questions (MCQs)

{By Pin2}

Objective Questions

1. **Binary search algorithem cannot be applied to _____ .**

    1. sortrd linked list
    2. sorted binary trees
    3. sorted linear array
    4. None of the given

2. **a * (b+c)-d is an example of _____ expression.**

    1. infix
    2. prefix
    3. postfix
    4. allfix

3. **Which of the following possible operation are performed on Table ADT?**

    1. Insert, Remove
    2. Find, Remove
    3. Insert, Find
    4. Insert, Find, Remove

4. **Following is a keyword of C++ _____ .**

    1. del
    2. delete
    3. Remove
    4. eliminate

5. **Which one of the following calling methods does not change the original value of the argument in the calling function?**

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1. Call by passing the value of the argument
2. Call by passing reference of the argument
3. Call by passing the address of the argument
4. Call by passing pointer of the argument

6. **Insertion in a linked list can be done at**

    1. Front only
    2. Back only
    3. Somewhere in middle only
    4. Front, Back and somewhere in the middle

7. **Consider we have performed the following operations on stack of size 5.**
   Push(10);
   Push(20);
   Push(30);
   Pop();
   Pop();
   Push(40);

    1. 10
    2. 20
    3. 40
    4. 20

8. **Suppose there are 100 elements in an equivalence class, so initially there will be 100 trees. The collection of these trees is called**

    1. Cluster
    2. Class
    3. Forest
    4. Bunch

9. **A queue where the dequeue operation does not depend upon FIFO, is called:**

    1. enqueue
    2. simple queue
    3. stack
    4. priority queue

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10. Array cells are _____ in computer memory.

1. Contiguous
2. Random
3. Store in multiple variables
4. Store in multiple functions

11. During the union by size method, all the array element initialized to -1 shows:

1. Tree is complete now
2. Initial condition of tree
3. Every tree has two elements each
4. None of given

12. next() method of List class is used to:

1. Moves current position backward one element
2. Moves the "current" pointer to two steps after the last elementof the array
3. Moves the current position forward one element
4. Moves the "current" pointer to two steps before the last element of the array

13. Which of the following heap method lowers the value of key at position 'p' by the amount 'delta'?

1. increaseKey(p,delta)
2. decreaseKey(p,delta)
3. percolateDown(p,delta)
4. remove(p,delta)

14. Binary Search can be categorized into which of following?

1. Greedy algorithm
2. Dynamic programming
3. Divide and conquer
4. Brute Force technique

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15. Suppose there is an image of 7 * 7, now we will have matrix of _____ rows and _____ columns.

1. 7, 7
2. 49, 49
3. 100, 100
4. 8, 8

16. AVL tree is linear data structure.

1. TRUE
2. FALSE
3. In some cases
4. None of the given

17. Parameters in function call are passed using,

1. Stack
2. Queue
3. Binary Search Tree
4. AVL Tree

18. If Ahmad is cousin of Ali and Ali is cousin of Asad then Ahmad is also cousin of Asad. This statement has the following property

1. Reflexivity
2. Symmetry
3. Transitivity
4. All of the given

19. The binary tree is the maximum level of its leaves (also called the depth).

1. Level
2. Width
3. Height
4. None of the given

20. ~BinarySearchTree() is a _____ .

1. Constructor

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2. Destructor
3. Switch case
4. Template method call

21. A hash function returns a _____ value.

1. Integer
2. Double
3. Float
4. Char

22. _____ only removes items in reverse order as they were entered.

1. Stack
2. Queue
3. Both of the given
4. None of the given

23. An array is a group of _____ memory locations.

1. Scattered
2. Isolated
3. Random (non-consecutive)
4. Consecutive

24. In singly linked list which node will keep track of starting position of the list.

1. Next Node
2. Previous Node
3. Head Node
4. Last Node

25. Leaf node of binary search tree contains _____.

1. One Null pointer
2. Three Null pointers
3. Two Null pointers
4. All of the given

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26. If we singly linked list to implement list, then there is an issue that it gives difficulty when we:

1. Move forward in the list
2. Move backward in the list
3. We will increase its size
4. We will decrease its size

27. A Linear Data Structure is the data structure in which data elements are arranged in a sequence or a linear list. Which of the following is Non Linear Data Structure?

1. Arrays
2. Linked Lists
3. Binary Search Trees
4. Stack

28. The expression DE+H* is called _____.

1. Prefix expression
2. Infixexepression
3. Postfix expression
4. Hybrid expression

29. In simple or singly linked list there is/are _____ pointer/s in each node.

1. One
2. Two
3. Three
4. Four

30. Which operation of queue data structure is used to get front element from the queue and then remove it from the queue?

1. enqueue()
2. dequeue()
3. front()
4. remove()

31. The expression AB+C* is called?

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1. Prefix expression
2. Postfix expression
3. Infix expression
4. Prefix and Infix expression

32. Which of the following operations returns "most recently entered value" from the stack?

1. Push
2. Recent
3. Top
4. First

33. Stack and Queue can be implemented using _____.

1. Singly Link List
2. Binary Tree
3. Binary Search Tree
4. AVL Tree

34. Which one of the following is known as "Last-In, First-Out" or LIFO Data Structure?

1. Linked List
2. Stack
3. Queue
4. Tree

35. The expression
if (! heap ->isEmpty())
checks

1. Heap is empty
2. Heap is full
3. Heap is not empty
4. Not a valid expression

36. Which of the following is correct statement?

1. An AVL tree is identical to BST expect height of the left and right subtree can differ by at most 1.

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2. An AVL tree is identical to BST expect height of the left and right subtree can differ by at least 1.
3. An AVL tree is identical to BST expect height of the left and right subtree must differ by at least 1.
4. An AVL tree is not identical to BST, its another kind of tree.

37. What will be the value of root of an AVL and BST if built from the same data?

1. Same
2. Different in some cases
3. Root of AVL is square of root of BST
4. None of the given

38. int htdiff = height(root->getLeft()) _____ height(root->getRight()); The above line of code is taken from AVL insert method. Complete it by selecting an appropriate symbol.

1. -
2. +
3. /
4. *

39. back() method of list class is used to:

1. Moves the "current" pointer to backward one element.
2. Moves the "current" pointer to backward two element.
3. Moves the "current" pointer to backward three element.
4. Moves the "current" pointer to backward four element.

40. Avl tree takes maximum _____ time to search an element.

1. 1.44 Log2n
2. Log2(n+n)
3. Log2(n+1)+1
4. 1.88 Log2n

41. Each node in singly link list has,

1. 1 pointer

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2. 2 pointers
3. 3 pointers
4. 4 pointers

42. If we use array to implement list, then there is an issue that it gives difficulty when:

1. We will access values randomly
2. We will remove data from it
3. We will increase its size
4. We will decrease its size

43. If there are N external nodes in a binary tree then what will be the no. of internal nodes in this binary tree?

1. N -1
2. N+1
3. N+2
4. N

44. Which of the following is NOT an open addressing technique to resolve collisions?

1. Quadratic probing
2. Double hashing
3. Cubic probing
4. Linear probing

45. Consider the following expression:
x - y * a + b / c
Which of the following is a correct equivalent expression for the above?

1. x y -a * b +c /
2. x *y a - b c / +
3. x y a * - b c / +
4. x y a * - b/ + c

46. Every AVL is _____.

1. Ternary Tree
2. Complete Binary Tree

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3. Heap
4. Binary Search Tree

47. New items are added at the _____ of the stack.

1. Bottom
2. Middle
3. Top
4. Center

48. What will be the postfix expression of following infix expression?
A*B/C+D-E

1. A B / C * D + E -
2. A B * C / D + E -
3. A B * C + D E / -
4. A B + C D / + E -

49. Binary Tree traversal can be performed with the help of _____ .

1. Recursive calls only
2. Non-recursive calls
3. Both Recursive and Non-recursive calls
4. None of the given

50. Which one of the following is NOT the property of equivalence relation?

1. Reflexive
2. Symmetric
3. Transitive
4. Associative

51. Sorting procedure normally takes _____ time.

1. NLogN
2. 2N
3. N*N*N
4. N

52. Suppose we have been given the following data set for a Queue.
7 5 2 4

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What will be the resultant Queue if we call enqueue(3) method? Note that 7 is the front element whereas 4 is rear element of queue.

1. 7 5 2 4
2. 3 7 5 2 4
3. 7 5 2 4 3
4. 5 2 4 3

53. In doubly linked list there is/are _____ pointer/s in each node.

1. One
2. Two
3. Three
4. Four

54. Left, right, info, and parent are the operation of _____ data structure.

1. Stack
2. Tree
3. Queue
4. Linked List

55. In C++, we place the class interface in _____ file.

1. .cpp
2. .cppp
3. .h
4. .hh

56. Suppose n is the number of nodes in a complete Binary Tree, then maximum steps required for a search operation are

1. $Log_2 (n+1) -1$
2. $Log_2 (n+1)$
3. $Log_2 (n) -1$
4. $Log_2 (n)$

57. The union operation is based on size or weight but the reducing the in-between links or path compression from _____ to the _____ is done by the find method.

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1. roots, node
2. nodes, root
3. root, root
4. node, node

58. If we have 1000 sets each containing a single different person. Which of the following relation will be true on each set:

1. Reflexive
2. Symmetric
3. Transitive
4. Associative

59. Consider the following infix expression.
7/8 + 9
If one converts the above expression into postfix. What would be the resultant expression?

1. 7 8 9 / +
2. 7 8 / + 9
3. / 7 8 + 9
4. 7 8 / 9 +

60. A table consists of several columns, known as

1. Entity
2. Tuple
3. Fields
4. Record

61. Heap can be used to implement

1. Stack
2. Linked list
3. Queue
4. Priority Queue

62. For reference variables, _____ sign is used.

1. ampersand
2. asterisk

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3. sigma
4. dollar

63. What will be result of following postfix expression?

1 2 3 * + 2 -

1. 3
2. 4
3. 5
4. 10

64. In _____, a programmer uses two pointers in the node, i.e. one to point to next node and the other to point to the previous node.

1. Linked list
2. Doubly-linked list
3. Array
4. Structure

65. The most difficult case for deleting a node from Binary Search Tree is when the node to be deleted _____.

1. is Leaf node
2. has only left children (subtree)
3. has only right children (subtree)
4. has both the left and right children (subtree)

66. What is the depth of any tree if the union operation is performed by height?

1. O(N)
2. O(N log N)
3. O(log N)
4. O(M log N)

67. Can we store elements with different data types in a single array?

1. Yes
2. No
3. In some cases

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4. None of the given

68. The main use of AVL tree is:

1. Searching of data
2. Storing of data
3. Insertion of data
4. Security of data

69. If we want to find 3rd minimum element from an array of elements, then after applying buildHeap method, how many times deleteMin method will be called?

1. 1
2. 2
3. 3
4. 4

70. For a perfect binary tree of height h, having N nodes, the sum of hights of nodes ia _____.

1. N - h - 1
2. N - 1
3. N - 1 + h
4. N - (h - 1)

71. If ahmad is boss of ehsan and ehsan is boss of umer then ahmad is also boss of umer.
The above mentioned relation is _____.

1. Reflexive
2. Symmetry
3. Transitive
4. None of the given

72. If unions are done by weight (size), the depth of any element is never greater than

1. log 3n
2. $\log_2 n$
3. $n \log_2 n$

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4. log n*n

73. Consider the linked list having data [6, 72, 35, 65, 25] stored in it. While current pointer is pointing to memory location having 72 stored in it. After calling add(4) function on the following linked list current point will point to memory location having value?

1. 72
2. 35
3. 4
4. 25

74. Select the one FALSE statement about binary trees:

1. Every binary tree has at least one node.
2. Every non-empty tree has exactly one root node.
3. Every node has at most two children.
4. Every non-root node has exactly one parent.

75. _____ is used for Reference variables in C++.

1. !
2. @
3. #
4. &

76. The total number of nodes on 10th level of a perfect binary tree are :

1. 256
2. 512
3. 1024
4. Can't be determined

77. One difference between a queue and a stack is

1. Queues require dynamic memory, but stacks do not.
2. Stacks require dynamic memory, but queues do not.
3. Queues use two ends of the structure, stacks use only one.
4. Stacks use two ends of the structure, queues use only one.

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78. When an executable program runs, it is loaded in the computer memory and becomes a _____ .

1. Thread
2. .h file
3. Process
4. None of the given

79. Consider the function X as under

```
int X (int& Value)
{
return Value;
}
```

Now a and b are integers in a calling function. Which one of the following is a valid call to the above function X?

1. a = X(b);
2. a = X(&b);
3. a = X(*b);
4. a = X(&*b);

80. For searching a particular number in Binary Search Tree (if it is not present), the maximum number of comparisons will be _____ comparison at each level.

1. 1
2. 2
3. 3
4. 4

81. If there are 56 internal nodes in a binary tree then how many external nodes this binary tree will have?

1. 54
2. 57
3. 56
4. 55

82. for every process executing, the last part of the memory is for _____ of the program.

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1. Data
2. Code
3. Stack
4. Heap

83. An efficient program executes faster and helps in _____ the usage of resources like memory and disk.

1. Maximizing
2. Minimizing
3. Equalizing
4. None of the given

84. Suppose a hash function returns 6 for the given value. At which indexof array the value will be saved?

1. 0
2. 5
3. 6
4. 7

85. Which property of equivalence relation is satisfied if we say: Ahmad is cousin of Ali and Ali is also cousin of Ahmad

1. Reflexivity
2. Symmetry
3. Transitivity
4. All of the given

86. The expression
if (! heap ->isFull())
check

1. Heap is not empty
2. Heap is not full
3. Heap is empty
4. Heap is full

87. _____ is an area in computer memory that is allocated dynamically.

1. Heap

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2. Stack
3. Queue
4. Linked List

88. If class A defines class B as its friend, then:

1. Class A can access private members of Class B
2. Class B can access only the public members of Class A
3. Class A can access only the public members of Class B
4. Class B can access private members of Class A

89. Non-recursive calls are faster than which of the following calls?

1. Parameterized
2. Recursive
3. Function
4. Non-Function

90. If there are N internal nodes in a binary tree then what will be the no. of external nodes in this binary tree?

1. N -1
2. N
3. N +1
4. N +2

91. The next field in the last node of a singly-linked list is set to _____.

1. NAN
2. 1
3. 'NULL
4. -1

92. If a tree has 20 edges/links, then the total number of nodes in the tree will be:

1. 19
2. 20
3. 21
4. Can't be determined

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93. The simplest case in a BST to delete a node is:

1. When the node, that is to be deleted is root node
2. When the node, that is to be deleted has both left and right child
3. When the node, that is to be deleted has only one child
4. When the node, that is to be deleted is a leaf node

94. Here is a small function definition:

```
void f(int i, int &k)
{
i = 1;
k = 2;
}
```

Suppose that a main program has two integer variables x and y, which are given the value 0. Then the main program calls f(x,y); What are the values of x and y after the function f finishes?

1. Both x and y are still 0.
2. x is now 1, but y is still 0.
3. x is still 0, but y is now 2.
4. x is now 1, and y is now 2.

95. Allocating and de-allocating memory for linked list nodes take _____ time than pre-allocated array.

1. Less
2. More
3. Equal
4. No

96. The percolateDown procedure will move the smaller value _____ and bigger value _____.

1. left, right
2. right, left
3. down, up
4. up, down

97. "new int[11]" will allocate memory for _____ integers.

1. 10

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2. 11
3. 12
4. 13

98. Which of the following properties are satisfied by Equivalence relationship?

1. Reflexive, Symmetric
2. Reflexive, Transitive
3. Symmetric, transitive
4. Reflexive, Symmetric and Transitive

99. If a hash function returns 4 for more than one value, this condition is called _____.

1. Merging
2. Collision
3. Mapping
4. Clustering

100. During union by size method, which data structure is used to improve the balancing of tree?

1. Array
2. Stack
3. Linked List
4. Tree

101. What will be the result of evaluation following expression? 5+3*2/(6-3)

1. 3
2. 5
3. 7
4. 10

102. Which of the following is correct about AVL Tree?

1. It is identical to BST except height of the left and right subtrees can differ by at least 1.

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2. It is identical to BST except height of the left and right subtrees must differ by at least 1.
3. It is not identical to BST, it is totally different kind of tree.
4. It is identical to BST except height of the left and right subtrees can differ by at most 1.

103. A complete binary tree having "N" nodes consists of _____ Levels.

1. $Log_2 (N+1) -1$
2. $Log_2 (N-1) -1$
3. $Log_2 (N+1) +1$
4. $Log_2 (N-1) +1$

104. In the call by _____ methodology, a copy of the object is passed to the called function.

1. Reference
2. Value
3. Reference & Value
4. Copy of the object can not be passed

105. Binary Search Tree voilates the condition of AVL tree when any node has balance equal to

1. 2 or -2
2. 1 or -1
3. 0
4. None of the options.

106. If there are N elements in an array then the number of maximum steps needed to find an element using Binary Search is _____.

1. N
2. N2
3. $Nlog_2N$
4. log_2N

107. If we want to find median of 50 elements, then after applying buildHeap methd, how many times deleteMin method will be called?

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1. 5
2. 25
3. 35
4. 50

108. The tree data structure is a

1. Linear data structure
2. Non-linear data structure
3. Graphical data structure
4. Data structure like queue

109. What will be postfix expression of the following infix expression?
Infix Expression: a+b*c-d

1. ab+c*d-
2. abc*+d-
3. abcd+*-
4. abc+*d-

110. If there are 100 elements in an equivalence class then we will have _____ sets initially.

1. 50
2. 100
3. 1000
4. 80

111. If the height of a perfect binary tree is 4. What will be the total number of nodes in it?

1. 15
2. 16
3. 31
4. 32

112. A binary tree of N nodes has _____ .

1. Log_{10} N levels
2. Log_2 N levels

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3. N / 2 levels
4. N x 2 levels

113. Which of the following statement is NOT correct regarding Table ADT?

1. In a table, the type of information in columns may be different.
2. A table consists of deveral columns known as entities
3. The row of a table is called a record
4. A major use of table is in databases where we build and use tables for keeping information

114. Which of the following can be used to reverse a string value?

1. Stack
2. Queue
3. Both of these
4. None of the given

115. Which of the following is NOT true regarding the maze generation?

1. Randomly remove walls until the entrance and exit cells are in same set
2. Do not remove a randomly chosen wall if the cells it separates are already in the same set
3. Remove a randomly chosen wall if the cells it separates are already in the same set
4. Removing a wall is the same as doing a union operation

116. In level-order traversal for Binary Search Tree, at each level, we visit the nodes in _____ order.

1. right-to-left
2. left-to-right
3. top to botton
4. any

117. Suppose there are a set of fruits and a set of vegetables. Both sets are _____ sets.

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1. Disjoint
2. Subsets
3. Whole
4. Equal

118. Local variables defined inside function body are _____ automatically at the end of function execution.

1. created
2. destroyed
3. incremented
4. decremented

119. _____ in AVL is logarithmic.

1. Updating
2. Searching
3. Deletion
4. Insertion

120. _____ is when function is calling to itself.

1. Loop
2. Recursion
3. Iteration
4. Nasted loop

121. The tree structure is a

1. Linear data structure
2. Non-linear data structure
3. Graphical data structure
4. data structure like queue

122. Suppose there is an image segmented into pixels. Each pixel has _____ neighbour(s).

1. 0
2. 4
3. 8

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4. 16

123. The _____ of a node in a binary tree is defined as the height of its left subtree minus height of its right subtree.

1. Height
2. Balance
3. Width
4. None of the above

124. The worst case of searching in binary search tree (BST) is:

1. When the data inserted in BST is sorted
2. When the height of left sub-tree is greater than right sub-tree
3. When the height of right sub-tree is greater than left sub-tree
4. When the tree is balanced

125. Y = &x[0];
In the above statement, we get address of the first location of the array x and store
it in y. Here "y" is:

1. rvalue
2. xvalue
3. lvalue
4. zvalue

126. Consider the following pseudo code

```
declare a stack of characters
while ( there are more characters in the word to read )
{
read a character
push the character on the stack
}
while ( the stack is not empty )
{
pop a character off the stack
write the character to the screen
}
```

What is written to the screen for the input "apples"?

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1. selpa
2. selppa
3. apples
4. aaappppplleess

127. Linked list always contains elements that can be described as,

1. Redundant
2. Recursive
3. Self-referential
4. Bidirectional

128. Which of the following is true regarding the maze generation?

1. Randomly remove walls until the entrance and exit cells are in same set.
2. Removing a wall is the same as doing a union operation
3. Do not remove a randomly chosen wall if the cells it separates are already in the same set
4. All of the given

129. The easiest case of deleting a node from BST is the case in which the node to be deleted _____.

1. Is a leaf node
2. Has left subtree only
3. Has right subtree only
4. Has both left and right subtree

130. The total number of nodes on 5th level of a perfect binary tree are :

1. 16
2. 15
3. 31
4. 32

131. The depth of a binary tree is

1. Total number of nodes in the tree
2. Number of leaf nodes in the tree

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3. Number of non-leaf nodes in the tree
4. Maximum level of a leaf

132. Write in postfix form 5 * (9 - 7)

1. 5 (9 7 -) *
2. 5 9 * 7 -
3. 5 9 7 - *
4. 5 (9 7 -)

133. What will be postfix expression of the following infix expression?
Infix Expression: a+b*c-d

1. ab+c*d-
2. abc*+d-
3. abc+*d-
4. abcd+*-

134. What are the basic things associated with data structures?

1. Space for each data item it stores
2. Time to perform each basic operation
3. Programming effort
4. All of the given

135. The _____ method of list data structure removes the element residing at the current position

1. Add
2. Next
3. Remove
4. Find

136. Every AVL tree is a binary search tree.

1. TRUE
2. FALSE
3. Not in some cases
4. None of the given

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137. In a tree, we link the nodes in such a way that it _____ a linear structure.

1. does not remain
2. forms
3. reverses
4. remains

138. In singly linked list a node consists of two parts:

1. Object and structure
2. Two pointers
3. Two objects
4. Object and pointer

139. Which property of equivalence relation is satisfied if we say: Ahmad R(is related to) Ahmad

1. Reflexivity
2. Symmetry
3. Transitivity
4. All of the given

140. Searching an element in an AVL tree takes maximum _____ time (where n is number of nodes in AVL tree).

1. $Log_2(n+1)$
2. $Log_2(n+1)$ -1
3. 1.44 Log_2n
4. 1.66 Log_2n

141. What is the maximum depth of recursive calls a function may make?

1. 1
2. 2
3. n (where n is the argument)
4. There is no fixed maximum

142. When we compare recursive method calls and non-recursive method calls, following statement is true.

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1. Recursion is implementted in the same way as other functioncalls are implemented
2. Non-recursive methods are always efficient than recursive methods
3. Both options are true
4. None of the given options are true

143. What will be the postfix expression of following infix expression? D+E*F/G

1. DE*F/G+
2. DE+F*G/
3. DEF*G/+
4. DE+FG*/

144. While implementing non-recusive traversal for Binary SearchTree, we need to implement _____ .

1. Queue
2. Stack
3. Min heap
4. Max heap

145. If a tree has 50 nodes, then the total edges/links in the tree will be:

1. 55
2. 51
3. 50
4. 49

146. Which of the following operation returns but do not removes top value of the stack?

1. push
2. pop
3. top
4. first

147. What is the time complexity of binary search with iteration?

1. O(nlogn)

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2. O(n2)
3. O(n)
4. O(logn)

148. Doubly Linked List always has _____ NULL pointer/s in a node.

1. One
2. Two
3. Three
4. Four

149. For compiler, a postfix expression is easier to evaluate than infix expression?

1. True
2. False

150. The new operation in C++ for dynamically allocating memory returns,

1. Size of the memory it has allocated
2. Pointer to the memory it has allocated
3. Both size and pointer to the memory it has allocated
4. Value of the memory it has allocated

151. Longest path from root node to farthest leaf node is called _____ of tree.

1. Level
2. Length
3. Depth
4. Node level

152. The variables which are destroyed automatically when a function's execution ends are:

1. Global variables
2. Local variables defined inside function body
3. Variables (objects) defined inside function body dynamically
4. Variables (objects) defined inside function body statically

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153. The difference between a "Binary Tree (BT)" and a "Binary Search Tree (BST)" is that,

1. A BST has two children per node whereas a BT can have none, one or two children per node
2. In BST nodes are inserted based on the values they contain
3. In BT nodes are inserted based on the values they contain
4. There is no difference

154. which of the following function don't belongs to the stack class?

1. push()
2. pop()
3. crash()
4. top()

155. Which of the following is not a form of exoression?

1. Infix
2. Postfix
3. Prefix
4. Pastfix

156. Which of the following statement is false?

1. Arrays are dense lists and static data structure
2. data elements in linked list need not be stored in adjecent space in memory
3. pointers store the next data element of a list
4. linked lists are collection of the nodes that contain information part and next pointer

157. _____ is the stack characteristic but _____ was implemented because of the size limitation of the array.

1. isFull(), isEmpty()
2. pop(), push()
3. isEmpty(), isFull()
4. push(), pop()

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158. In 1990, Bill pugh proposed an enhancement on linked lists and the new data structure was termed as

1. Linked list
2. B-Tree
3. Skip list
4. Spelling checker

159. Each operator in a postfix expression refers to the previous _____ operand(s).

1. One
2. Two
3. Three
4. Four

160. In which of the following function signatures, the value of variable "num" cannot be changed in function body?

1. int cube(int num)
2. int cube(int& num)
3. int cube(const int& num)
4. int cube(int* num)

161. Factorial is an example of _____ function.

1. Recusive
2. Non-Recursive
3. Cube
4. Log

162. In simple implementation of stack, isFull() method is used due to _____.

1. Limitation of array
2. Strength of array
3. Linked list connectivity
4. Complexity of linked list

163. To search an element in ALV tree, it takes maximum 1.88 Log2n time.

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1. TRUE
2. FALSE
3. In some cases
4. None of the given

164. Each node in Binary Search Tree has

1. 1 pointer
2. 2 pointers
3. 3 pointers
4. 4 pointers

165. A binary tree with 24 internal nodes has _____ external nodes.

1. 22
2. 23
3. 48
4. 25

166. The array in binary search is sub divided _____.

1. Once
2. Twice
3. N time
4. Untill a sublist is no more divisible

167. There are four cases of rotation is an _____ tree.

1. ELV
2. EVL
3. AVL
4. ALV

168. For a perfect binary tree of height 4, What will be the sum of hights of nodes?

1. 31
2. 30
3. 27
4. 26

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169. Which one of the following is NOT true regarding the skip list?

1. Each list Si contains the special keys + infinity and -infinity
2. List S0 contains the keys of S in non-decreasing order
3. List Sh contains only the n special keys
4. Each list is a subsequence of previous one

170. Every AVL is _____.

1. Binary Tree
2. Complete Binary Tree
3. Binary Search Tree
4. None of the given

171. A solution is said to be efficient if it solves the problem within its resource constraints i.e. hardware and time.

1. True
2. False

172. Each node in doubly link list has

1. 1 pointer
2. 2 pointers
3. 3 pointers
4. 4 pointers

173. If there are 100 elements in a heap, and 100 deleteMin operation are performed, will get _____ list.

1. Sorted
2. Unsorted
3. Nonlinear
4. None of the given

174. In case of insertion of right inner node in BST,

1. we need to apply single left rotation to make it AVL tree.
2. we need to apply single right rotation to make it AVL tree.
3. single left rotation first and then single right rotation to make it AVL tree.

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4. single right rotation first and then single left rotation to make it AVL tree.

175. What is the hash function used in linear probing?

1. hi(x)=hash(x) mod table size
2. hi(x)=(hash(x) + f(i^2)) mod table size
3. hi(x)= (hash(x)+ f(i)) mod table size
4. hi(x)= X mod 17

176. In tree, the search operation is _____ as compared to the linked list.

1. Very fast
2. not fast
3. equally time-taken
4. very slow

177. If there are 23 external nodes in a binary tree then what will be the no. of internal nodes in this binary tree?

1. 23
2. 24
3. 22
4. 21

178. given the values are the array representation of heap:
12 23 26 31 34 44 56 64 78 100
If we perform 4 deleteMin operations, the last element deleted is _____.

1. 31
2. 34
3. 44
4. 56

179. Recursive call of a function use _____ data structure.

1. Linked List
2. Queue
3. Stack

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4. Table

180. Which of the following statement is NOT true for reference variable?

1. Once a reference is created, it cannot be later made to reference another object.
2. Reference cannot be NULL
3. References can be uninitialized.
4. It is not possible to refer directly to a reference object after it is defined.

181. For Binary Search Tree, we call the findMax() method as _____.

1. findMin(tree->getLeft())
2. findMin(tree->getRight())
3. findMin(tree->getCenter())
4. findMin(tree->getDown())

182. An array x of 100 integers is declared as

1. int x = [100];
2. int x[100];
3. integer array x = 100;
4. integer x [] = 100;

183. given the values are the array representation of heap:
12 23 26 31 34 44 56 64 78 100
What is the 5th smallest element in the given heap?

1. 31
2. 34
3. 44
4. 56

184. Trying to remove an element from an empty stack is called _____
.

1. Garbage collector
2. Overflow of stack
3. Empty collection

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4. Underflow of stack

185. Consider the following function:

```
void test_a(int n)
{
cout << n << " ";
if (n>0)
test_a(n-2);
}
```

What is printed by the call test_a(4)?

1. 4 2 0
2. 0 2 4
3. 0 2
4. 2 4

186. The balance of a node is the result of "height of left subtree" _____ "height of right subtree".

1. Plus
2. Minus
3. Multiply
4. Divided by

187. From Operating System point of view, the recursive function calls are made with the help of _____.

1. Queue
2. Stack
3. Linked list
4. Binary Search Tree

188. get(?) method of list class is used to:

1. Get element from the last position
2. Get element from the first position
3. Get element from the middle position
4. Get element at the given position

189. Consider a hash table of size seven, with starting index zero, and a hash function (3x + 4)mod7. Assuming the hash table is initially empty,

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which of the following is the contents of table when the sequence 1, 3, 8, 10 is inserted into the table using closed hashing? NOte that '_' denotes an empty location in the table.

1. 8, _, _, _, _, _, 10
2. 1, _, _, _, _, _, 3
3. 1, 10, 8, _, _, _, 3
4. 1, 8, 10, _, _, _, 3

190. Which of the is NOT true regarding the maze generation?

1. Randomly remove walls until the entrance and exit cells are in same set.
2. Removing a wall is the same as doing a union operation
3. Remove a randomly chosen wall if the cells it separates are already in same set
4. Do not remove a randomly chosen wall if the cells it separates are already in same set.

191. Which of the following applications may use a stack?

1. Accessing a shadred resource
2. Parentheses balancing program
3. Buffering messages
4. Waiting list

192. In the statement int& a=b;

1. a and b pointing to two different memory location
2. a and b are two different names of the same memory location
3. a and b are two different variable names
4. b hold the address of variable a

193. In the poat-order traversal of a binary search tree, nodes process as:

1. Left-subtree, Right-subtree, Root
2. Rightt-subtree, Root, Left-subtree
3. Left-subtree, Root, Right-subtree
4. Rightt-subtree, Left-subtree, Root

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194. Which one of the following statement is correct?

1. Array size is fixed once it is created
2. Link List size is fixed once it is created
3. Binary Search Tree size is fixed once it is created
4. AVL Tree size is fixed once it is created

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